TD8 : WORKSTATION STUDY - TIME ANALYSIS

Exercise 1: Chrono-analysis

Time observed (To)

The timings made it possible to determine the times observed To.

The person observed was working at a pace that is not necessarily representative of their usual work. The stress of the timing, the desire to show that she is a productive person, ... distort the timed result. The observed times must be corrected according to these parameters.

The observed times do not take into account the working conditions, the experience of the people observed, ... The observed times will also be corrected by taking into account these different factors.

Determination of normal time (Tn)

The observed operator works at a pace that may be different from that of a skilled and experienced operator working at a normal pace called 100 pace.

Tn = $\frac{To \times operator's\ pace\ (\%)}{100}$

Determination of the standard time (Ts)

The normal time is increased in order to take into account the stops of the operator for personal needs, the fatigue of the operator, the hazards of the production, ... This standard time will be used to determine the rates, the efficiency of the operator, …

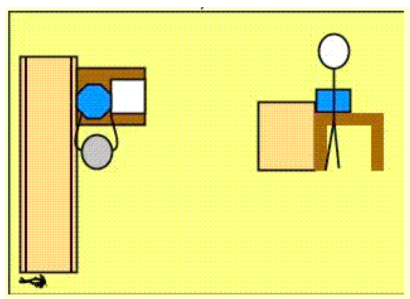
The markup changes according to the position of the operator

Ts = Tn x (1+ markup%)

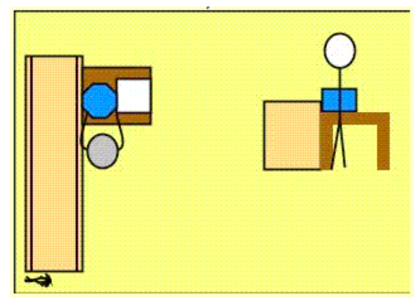
Exercise

You have to carry out a chrono analysis on a control station. An operator is assigned to this workstation.
Here are the identified sequences, top view and side view:

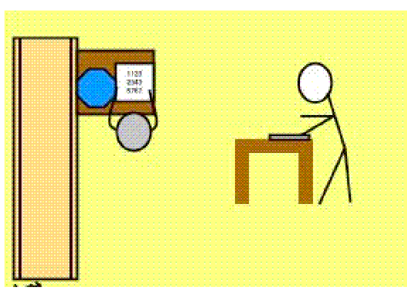
1- Take the part



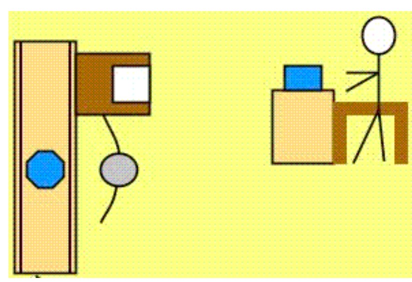
2- Measure



3- Report dimensions



4- Put the part on the conveyor



Here are the results in dmh of the timings of an operator who handles a part of 2 kg, puts it on a table and controls it, then transfers the values on a sheet.

SEQUENCES	Time observed (dmh)				
	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5
seize the part	15	13	18	16	17
measure	56	63	60	62	57
report dimensions	45	40	48	45	42
put the part on the conveyor	18	20	23	23	20

Here is the table of pace judgments estimated by the observer:

SEQUENCES	Operator's pace (%)				
	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5
seize the part	100	115	90	100	90
measure	110	100	105	100	110
report dimensions	90	100	85	90	95
put the part on the conveyor	100	100	90	90	100

Determine the normal times for each sequence and observation.

Determine the representative normal time for each sequence. The median time will be used.

The times will be given in dmh and rounded to the nearest whole number.

SEQUENCES	Normal time (dmh)				
	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5
seize the part					
measure					
report dimensions					
put the part on the conveyor					

SEQUENCES	Normal time (dmh)
seize the part	
measure	
report dimensions	
put the part on the conveyor	

Fictitious table of mark-up coefficient:

Operator's position	Transported mass (kg)		
	0 to 1	1 to 3	3 to 5
Standing	0%	8%	12%
Sitting	0%	10%	14%
Standing leaning	10%	15%	20%
Kneeling	15%	20%	30%

Determine the standard time for each sequence.

The times will be given in dmh and rounded to the nearest whole number.

SEQUENCES	Standard time (dmh)
seize the part	
measure	
report dimensions	
put the part on the conveyor	

Determine the hourly pace for this job (parts/hour).